

Full safe-packet frame-current Doob decomposition

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-078 defined the remaining coefficient-dominant object without assigning individual lower Wick chaoses to arbitrary Bony channels. Its definition is canonical: subtract the already paid nonresonant and payload-comparable orientations from the complete translated endpoint, then apply the causal projection once to the whole residual. R-078 proved a Doob bracket identity for factorised bilinear components, but deliberately left the exact full-packet reconstruction open.

This note supplies that reconstruction. The complete translated current is placed in one Hilbert space carrying the spatial integral, generator sum, direction sum, and the positive Q_{II} metric. Martingale orthogonality is then applied to the full terminal current, not term by term to its Taylor forest. A root-first insertion of the partial-control current produces three exact blocks: a low conditional-current term, a current-control injection term, and a future-feedback innovation term. Each shell retains its positive polarization square and its exact Wick covariance trace. The safe-packet identity then subtracts the two already paid orientations at both endpoints. Consequently both R-073 first variations, the terminal square, coefficient curvature, the complete R-063 forest, the R-066/R-070 trace, and the complete low safe endpoint remain exactly once.

The reconstruction is an algebraic advance, not the desired lower bound. Neither of the two shell blocks has a generic sign. The low term contains a conditional mean of the terminal future-controlled current and is not merely the R-066 finite-low boundary. The future innovation has both a coefficient channel and a derivative-feedback channel. A sharp one-root example remains negative after the positive current square and Wick trace are put back.

Two useful positive facts do survive. First, the controlled shells obey a genuinely spatially weighted Cameron–Martin square-function estimate with one use of the total control energy. Second, the uncontrolled or predictable translated base current has an exact same-point heat-semigroup projection, because future field values and derivatives are independent under the common even covariance. These facts identify the production-specific near/far continuation, but do not prove its far-tail estimate.

All exact identities are at finite cutoff, at a fixed positive production floor, with deterministic Q_{II} and deterministic covariance increments. The controls are strict-past predictable, mutually orthogonal, one-shot whole-shell controls. This note does not prove the production weighted packet lower bound, controlled-shell one-use for the energy, the $q = 10/9$ Nelson moment, a measure theorem, cutoff or floor removal, infinite volume, or Sector-A closure. The tier remains T4.

2 The complete current endpoint

Let $\ell = j_0 - 1$, $A^* = A^{(J)}$, and $A^0 = A^{(\ell)} = A_{<j_0}$. For a partial control H , define the complete Q_{II} -weighted current

$$\mathcal{J}(H)_{\alpha i}(x) = Q_{\text{II}}^{1/2} M_{\alpha}(U + H)^T \{D_i U + D_i H\}. \quad (2.1)$$

All generator and direction coordinates and the spatial L^2 integral are part of one real Hilbert space $\mathcal{H}$. Put

$$\mathcal{B}(H) = B(U + H), \quad (2.2)$$

and

$$\mathcal{W}_J(H) = \frac{1}{2} \|\mathcal{J}(H)\|_{\mathcal{H}}^2 - \frac{1}{2} \sum_i \int \mathcal{B}(H) : \Gamma_{J,i} dx. \quad (2.3)$$

The R-070 terminal identity gives, exactly,

$$\Delta V_J^{\text{ren}}(U; H) = \mathcal{W}_J(H) - \mathcal{W}_J(0). \quad (2.4)$$

Thus (2.3) is not a model replacement: it is the complete covariance-normal translated current endpoint already reached by R-066, R-069, and R-070.

Write

$$\mathcal{J}_* = \mathcal{J}(A^*), \quad \mathcal{J}_0 = \mathcal{J}(A^0), \quad \mathcal{J}_j = \mathcal{J}(A^{(j)}), \quad (2.5)$$

and use the same subscripts for $\mathcal{B}$. Let

$$P_j X = \mathbb{E}[X \mid \mathcal{F}_j], \quad d_j = P_j - P_{j-1}. \quad (2.6)$$

At finite cutoff P_J is the identity on every terminal current. Define the root-first increments

$$f_j = d_j(\mathcal{J}_j - \mathcal{J}_0), \quad i_j = d_j(\mathcal{J}_* - \mathcal{J}_j). \quad (2.7)$$

They obey

$$f_j + i_j = d_j(\mathcal{J}_* - \mathcal{J}_0), \quad d_j \mathcal{J}_j = d_j \mathcal{J}_0 + f_j. \quad (2.8)$$

The sequence f_j is not the martingale-difference sequence of one fixed terminal variable: its middle current $\mathcal{J}_j$ changes with j . This point prevents an automatic reuse of R-077 fresh-packet cancellation.

3 Exact root-first Doob decomposition

Assume the deterministic covariance split

$$\Gamma_J = \Gamma_\ell + \sum_{j=j_0}^J \Delta \Gamma_j. \quad (3.1)$$

Define the low conditional-current block

$$\begin{aligned} \mathfrak{L}_\ell &= \frac{1}{2} \mathbb{E} \{ \|P_\ell \mathcal{J}_*\|_{\mathcal{H}}^2 - \|P_\ell \mathcal{J}_0\|_{\mathcal{H}}^2 \} \\ &\quad - \frac{1}{2} \mathbb{E} \sum_i \int (\mathcal{B}_* - \mathcal{B}_0) : \Gamma_{\ell,i} dx. \end{aligned} \quad (3.2)$$

For every active root define the injected-current block

$$\begin{aligned} \mathfrak{F}_j &= \mathbb{E} \left\{ \langle d_j \mathcal{J}_0, f_j \rangle + \frac{1}{2} \|f_j\|_{\mathcal{H}}^2 \right\} \\ &\quad - \frac{1}{2} \mathbb{E} \sum_i \int (\mathcal{B}_j - \mathcal{B}_0) : \Delta \Gamma_{j,i} dx, \end{aligned} \quad (3.3)$$

and the future-feedback innovation block

$$\begin{aligned} \mathfrak{I}_j &= \mathbb{E} \left\{ \langle d_j \mathcal{J}_j, i_j \rangle + \frac{1}{2} \|i_j\|_{\mathcal{H}}^2 \right\} \\ &\quad - \frac{1}{2} \mathbb{E} \sum_i \int (\mathcal{B}_* - \mathcal{B}_j) : \Delta \Gamma_{j,i} dx. \end{aligned} \quad (3.4)$$

Theorem 3.1 (full-current identity). Under the hypotheses above,

$$\mathbb{E}\{\mathcal{W}_J(A^*) - \mathcal{W}_J(A^0)\} = \mathfrak{L}_\ell + \sum_{j=j_0}^J (\mathfrak{F}_j + \mathfrak{J}_j). \quad (3.5)$$

Proof. Hilbert martingale orthogonality gives, for $H = *$ and $H = 0$,

$$\mathbb{E}\|\mathcal{J}_H\|_{\mathcal{H}}^2 = \mathbb{E}\|P_\ell \mathcal{J}_H\|_{\mathcal{H}}^2 + \sum_{j=j_0}^J \mathbb{E}\|d_j \mathcal{J}_H\|_{\mathcal{H}}^2. \quad (3.6)$$

The difference of the j th squares is

$$\begin{aligned} & \langle d_j \mathcal{J}_0, f_j + i_j \rangle + \frac{1}{2} \|f_j + i_j\|^2 \\ &= \langle d_j \mathcal{J}_0, f_j \rangle + \frac{1}{2} \|f_j\|^2 + \langle d_j \mathcal{J}_j, i_j \rangle + \frac{1}{2} \|i_j\|^2. \end{aligned} \quad (3.7)$$

In particular, the easy-to-miss cross term $\langle f_j, i_j \rangle$ is contained in $\langle d_j \mathcal{J}_j, i_j \rangle$. The coefficient difference has the simultaneous split

$$\mathcal{B}_* - \mathcal{B}_0 = (\mathcal{B}_j - \mathcal{B}_0) + (\mathcal{B}_* - \mathcal{B}_j). \quad (3.8)$$

Insert (3.7) and (3.8) into (3.6), use (3.1), and compare with (2.3). This proves (3.5). No inequality or termwise forest assignment has entered. $\square$

The future block also has the exact polarization

$$\langle d_j \mathcal{J}_j, i_j \rangle + \frac{1}{2} \|i_j\|^2 = \frac{1}{2} \{\|d_j \mathcal{J}_*\|^2 - \|d_j \mathcal{J}_j\|^2\}. \quad (3.9)$$

Equation (3.9) is signless after the trace in (3.4) is restored. In particular, the trace is not generally the negative of the second square.

4 Exact reconstruction of the canonical safe packet

Let the already paid orientation be

$$\mathcal{R}_\leq(H) = N_{3,\text{nr}}(H) + \hat{T}_\leq(H). \quad (4.1)$$

R-078 defines

$$\mathfrak{B}_{J,\text{alg}}^>(H) = \Delta V_J^{\text{ren}}(U; H) - \mathcal{R}_\leq(H), \quad (4.2)$$

then applies its causal projection once to the whole bracket. Therefore

$$\mathbb{E}\mathfrak{B}_{J,\text{safe}}^> = \mathbb{E}\{\mathfrak{B}_{J,\text{alg}}^>(A^*) - \mathfrak{B}_{J,\text{alg}}^>(A^0)\}. \quad (4.3)$$

Combining (2.4), (3.5), and (4.2) proves the exact full-packet identity

$$\begin{aligned} \mathbb{E}\mathfrak{B}_{J,\text{safe}}^> &= \mathfrak{L}_\ell + \sum_{j=j_0}^J (\mathfrak{F}_j + \mathfrak{J}_j) \\ &\quad - \mathbb{E}\{\mathcal{R}_\leq(A^*) - \mathcal{R}_\leq(A^0)\}. \end{aligned} \quad (4.4)$$

This closes the algebraic reconstruction left open by R-078. It does not replace the right-hand side by a single future-coefficient bracket.

Two distinct low objects must be kept straight. First, reconstructing the complete terminal endpoint from the packet gives

$$\mathbb{E}\mathfrak{B}_{J,\text{alg}}^>(A^*) = \mathbb{E}\mathfrak{B}_{J,\text{alg}}^>(A^0) + \mathbb{E}\mathfrak{B}_{J,\text{safe}}^>. \quad (4.5)$$

The first term on the right is the complete low safe endpoint

$$\Delta V_J^{\text{ren}}(A^0) - N_{3,\text{nr}}(A^0) - \hat{T}_{\leq}(A^0), \quad (4.6)$$

not only the R-066 finite-low current boundary. Second, $\mathfrak{L}_\ell$ in (3.2) contains $P_\ell \mathcal{J}_*$ and hence a conditional mean of the entire future-controlled current. It is not presently bounded by the finite-low estimate for (4.6).

For every endpoint H , R-078 also gives the exact companion expansion

$$\mathfrak{B}_{J,\text{alg}}^>(H) = \mathcal{S}_{0,\Gamma}(H) + L(H) + N_2(H) + \hat{T}_{>}(H) + \mathcal{P}(H). \quad (4.7)$$

The first variation $\mathcal{S}_{0,\Gamma}$ contains both the restored R-073 Gaussian and control-current variations. Moreover,

$$\mathcal{P}(H) = \left\langle w, Q_{\text{II}}(E_2 + E_3)^T G \right\rangle + \frac{1}{2} |\Delta w + c|_{Q_{\text{II}}}^2 - \frac{1}{2} F_H : \Gamma_J. \quad (4.8)$$

Thus the projector-free square, coefficient curvature, and trace occur once. The coefficient term F_H means the complete R-063 $P_4/P_2/\Sigma Q/P_0$ and P_3/P_1 forest after reassembly; (4.4) never distributes those lower chaoses separately. This is why (4.4) preserves the R-063, R-066, R-070, and R-073 authorities simultaneously.

5 The future-control commutator has two channels

Set

$$R_j^{\text{f}} = A^* - A^{(j)}, \quad \Delta_j^{\text{f}} M = M(U + A^*) - M(U + A^{(j)}). \quad (5.1)$$

Direct expansion of (2.1) gives

$$\boxed{\mathcal{J}_* - \mathcal{J}_j = Q_{\text{II}}^{1/2} \{ (\Delta_j^{\text{f}} M)^T (DU + DA^{(j)}) + M(U + A^*)^T D R_j^{\text{f}} \}. \quad (5.2)}$$

Consequently $i_j = d_j(\mathcal{J}_* - \mathcal{J}_j)$ contains both a future coefficient innovation and a future derivative-control innovation. Strict-past predictability yields

$$R_j^{\text{f}} = \sum_{k>j} a_k, \quad d_j R_j^{\text{f}} = \sum_{k>j} d_j a_k, \quad (5.3)$$

but does not make (5.3) vanish. A reduction to coefficient innovation alone would delete the second line of (5.2).

The coefficient companion is likewise exact:

$$\begin{aligned} \mathcal{B}_* - \mathcal{B}_j = \sum_{\alpha} \{ & M_{\alpha,j} Q_{\text{II}} (\Delta_j^{\text{f}} M_{\alpha})^T + (\Delta_j^{\text{f}} M_{\alpha}) Q_{\text{II}} M_{\alpha,j}^T \\ & + (\Delta_j^{\text{f}} M_{\alpha}) Q_{\text{II}} (\Delta_j^{\text{f}} M_{\alpha})^T \}. \end{aligned} \quad (5.4)$$

The last line is coefficient curvature. Equations (3.9) and (5.4) show precisely why the future block must retain square and trace together.

6 Two exact positive tools

6.1 Spatially weighted Cameron–Martin square function

Write each one-shot control shell as $a_k = K_k h_k$, where a_k is $\mathcal{F}_{k-1}$ -measurable, supported in shell k , and

$$\|K_k\|_{\mathcal{H}_k \rightarrow L^2} \lesssim 2^{-2k}. \quad (6.1)$$

Then $d_j a_k = 0$ for $j \geq k$. Hilbert martingale orthogonality gives

$$\sum_{j < k} \mathbb{E} \|d_j a_k\|_2^2 \leq \mathbb{E} \|a_k\|_2^2. \quad (6.2)$$

Multiplying by 2^{4k} , using (6.1), and summing the mutually orthogonal spatial shells proves

$$\boxed{\sum_k \sum_{j < k} 2^{4k} \mathbb{E} \|d_j a_k\|_2^2 \lesssim \mathbb{E} \sum_k \|h_k\|_{\mathcal{H}_k}^2.} \quad (6.3)$$

Equivalently, with the standard shell norm,

$$\sum_j \mathbb{E} \|d_j A\|_{H^2}^2 \lesssim \mathbb{E} X. \quad (6.4)$$

This is a genuine weighted one-use result for the control coordinate. It is not yet a bound for the current bracket in (3.3)–(3.4), because the other factor has no proved compensating negative spatial regularity.

6.2 Predictable base-current heat projection

Let $U = U_j + V_{>j}$ and $DU = DU_j + DV_{>j}$. For a common real-even covariance,

$$\text{Cov}\{V_{>j}(x), D_i V_{>j}(x)\} = 0. \quad (6.5)$$

Joint Gaussianity makes the two variables independent at each x . If

$$\overline{M}_j = P_{\Sigma_{j+1:j}} M, \quad (6.6)$$

then the base current satisfies

$$\boxed{P_j \{M(U)^T D_i U\} = \overline{M}_j(U_j)^T D_i U_j.} \quad (6.7)$$

The same formula holds after an $\mathcal{F}_{j-1}$ -predictable translation, with U_j, DU_j translated by that control. Subtracting the $j-1$ formula gives

$$\begin{aligned} d_j \mathcal{J}_i &= \overline{M}_j(U_j)^T d_{j,i} \\ &\quad + \{\overline{M}_j(U_j) - \overline{M}_j(U_{j-1})\}^T D_i U_{j-1} \\ &\quad + \{\overline{M}_j - \overline{M}_{j-1}\}(U_{j-1})^T D_i U_{j-1}. \end{aligned} \quad (6.8)$$

These are respectively the derivative injection, centered value innovation, and heat compensator. Formula (6.7) does not apply to $\mathcal{J}_*$ when the terminal future feedback itself depends on root j .

7 Why the generic weighted shortcuts fail

7.1 Square-function Cauchy is exponent-supercritical

R-078 proves, for a raw-current order s ,

$$\|A^2 DA\|_{B_{1,1}^s} \lesssim X^{(1+s)/4} Y^{(5-s)/12}. \quad (7.1)$$

Spending (6.4) by Cauchy contributes $X^{1/2}$. The resulting powers are

$$a = \frac{3+s}{4}, \quad b = \frac{5-s}{12}, \quad c = 1 - a - b = -\frac{1+s}{6}. \quad (7.2)$$

At $s = 3/5$ this is

$$X^{9/10} Y^{11/30}, \quad c = -\frac{4}{15}. \quad (7.3)$$

There is no positive Young slack and hence no finite random Young remainder. The exact square, trace, forest, and paid subtraction must be used before Cauchy; (6.3) alone cannot close the packet.

7.2 Expected budgets do not imply predictable Carleson control

Fix a shell frequency N and an event $E \in \mathcal{F}_j$ of probability $p = N^{-6}$. On one normalized carrier set

$$h_{j+1} = p^{-1/2} \mathbf{1}_{E \in N}, \quad a_{j+1} = N^{-2} h_{j+1}. \quad (7.4)$$

Then, up to the fixed carrier normalization,

$$\mathbb{E} \|h_{j+1}\|^2 = 1, \quad \mathbb{E} \|a_{j+1}\|_6^6 = p \{N^{-2} p^{-1/2}\}^6 = 1. \quad (7.5)$$

But on E the conditional future energy is

$$\mathbb{E} [\|h_{j+1}\|^2 \mid \mathcal{F}_j] = p^{-1} = N^6. \quad (7.6)$$

Thus the two expected Boué–Dupuis budgets do not control an essential-sup martingale BMO or predictable Carleson norm.

7.3 An unweighted bracket has no generic spatial gain

On a one-step symmetric tree let $\varepsilon \in \{-1, 1\}$ be the centered Rademacher root, $\mathbb{E}\varepsilon = 0$, and take

$$H_N = \varepsilon \cos(Nx), \quad Y_N = -\varepsilon N^s \cos(Nx). \quad (7.7)$$

The L^2 square function of H_N and the homogeneous H^{-s} norm of Y_N stay bounded, whereas

$$\mathbb{E} \int H_N Y_N \, dx = -\frac{1}{2} \varepsilon^2 N^s. \quad (7.8)$$

Therefore the unweighted R-078 Doob identity cannot imply the missing spatially weighted bracket estimate by an abstract embedding.

8 Adapted Wick–carré-du-champ no-go

The positive innovation square also has no universal sign once its exact trace is restored. Let $\xi \sim N(0, 1)$, $H_2(\xi) = \xi^2 - 1$, $Q = \Gamma = 1$, and use the scalar frame $m(a) = 1 + a$. Allow a later value control $A = A(\xi)$ and set its spatial derivative control to zero. After subtracting the base term and all first variations, the exact remainder is

$$\mathcal{R}(A) = \frac{1}{2}A^2H_2 = \frac{1}{2}(A\xi)^2 - \frac{1}{2}A^2. \quad (8.1)$$

The two terms on the right are exactly the terminal current square and Wick trace.

For $A_\alpha = \alpha(\xi^2 - 4) = \alpha(H_2 - 3)$, probabilists' Hermites give

$$A_\alpha^2 = \alpha^2(H_4 - 2H_2 + 11), \quad (8.2)$$

$$A_\alpha^2H_2 = \alpha^2(H_6 + 6H_4 + 15H_2 - 4). \quad (8.3)$$

Consequently

$$\mathbb{E}\mathcal{R}(A_\alpha) = -2\alpha^2, \quad (8.4)$$

while

$$\frac{1}{2}\mathbb{E}(A_\alpha\xi)^2 = \frac{7}{2}\alpha^2, \quad -\frac{1}{2}\mathbb{E}A_\alpha^2 = -\frac{11}{2}\alpha^2. \quad (8.5)$$

Moreover

$$dA_\alpha = A_\alpha - \mathbb{E}A_\alpha = \alpha H_2, \quad \mathbb{E}|dA_\alpha|^2 = 2\alpha^2, \quad (8.6)$$

so the natural coefficient carré-du-champ occurs with exactly the wrong sign:

$$\boxed{\mathbb{E}\mathcal{R}(A_\alpha) = -\mathbb{E}|dA_\alpha|^2.} \quad (8.7)$$

This is not an omitted-chaos artefact: (8.3) displays the complete finite forest including its negative zero chaos.

There is also a bounded positive-frame fixture. For $a > 0$ take $A = a \exp(-\xi^2)$, so $m(A) \geq 1$. Direct Gaussian integration gives

$$\frac{1}{2}\mathbb{E}(A\xi)^2 = \frac{a^2}{10\sqrt{5}}, \quad -\frac{1}{2}\mathbb{E}A^2 = -\frac{a^2}{2\sqrt{5}}, \quad (8.8)$$

and hence

$$\mathbb{E}\mathcal{R}(A) = -\frac{2a^2}{5\sqrt{5}} < 0. \quad (8.9)$$

A diagonal two-component matrix lift leaves the first channel unchanged. Thus positive Q , a matrix-square coefficient, a bounded smooth adapted control, and the positive current square do not yield generic positivity.

This is an abstract adapted root counterexample, not a production $\mathfrak{B}^>$ counterexample. It omits the production frequency geometry, harmonic enhancement, exact paid subtraction, and the remaining signed companions. It forbids only a universal rootwise positivity or nonnegative carré-du-champ proof.

9 Conditional positive-gain ledger and successor

The failed routes identify a quantitative target. Suppose that, after the exact square–trace–forest re-assembly, the remaining production adapted coefficient obeys for some $\gamma > 0$

$$\|\mathbb{K}_\gamma(A)\|_{H^{-3/5+\gamma}} \leq R_\gamma \|A\|_{H^{2/5}}, \quad R_\gamma \in \bigcap_{p<\infty} L^p. \quad (9.1)$$

This is an assumption, not a theorem of R-063 for adapted controls. The standard interpolation gives

$$\|A\|_{H^{2/5}} \lesssim X^{1/10} Y^{2/15}. \quad (9.2)$$

Apply the R-078 payload estimate at $\beta = 3/5 - \gamma$. The resulting term has powers

$$R_\gamma X^{1/2-\gamma/4} Y^{1/2+\gamma/12}. \quad (9.3)$$

The exact Young ledger is

$$a = \frac{1}{2} - \frac{\gamma}{4}, \quad b = \frac{1}{2} + \frac{\gamma}{12}, \quad c = \frac{\gamma}{6}, \quad p_R = \frac{6}{\gamma}. \quad (9.4)$$

For the R-063-inspired target $\gamma = 3/10$,

$$(a, b, c) = \left(\frac{17}{40}, \frac{21}{40}, \frac{1}{20} \right), \quad p_R = 20, \quad (9.5)$$

with Young losses $\eta^{-17/2} \zeta^{-21/2}$. At $\gamma = 0$ the route is exactly critical. Thus any successful absolute near-block theorem must produce a genuine positive gain, or instead prove a direct signed production Schur inequality.

The safest continuation is now precise.

1. Split probability root j from spatial output shell m only after the exact identity (4.4).
2. In the far region $m \geq j + C$, prove a production current-tail estimate from the heat formula (6.7)–(6.8), and combine it with (6.3). No such tail is claimed here.
3. In the near region $|m - j| \leq C$, retain $\mathfrak{F}_j + \mathfrak{J}_j$ with both positive squares, both traces, the value-innovation and heat compensator, the complete R-063 forest, and the paid subtraction. Prove either (9.1) or a direct signed Schur bound.
4. Bound the conditional low current $\mathfrak{L}_\ell$ separately and then add the complete low endpoint (4.6). Neither may be renamed as the other.

10 Evidence map and route decisions

The exact dependency chain is

$$\begin{aligned} \text{R-063 forest} &\rightarrow \text{R-066 current/trace} \rightarrow \text{R-070 terminal current} \\ &\rightarrow \text{R-073 first variations} \rightarrow \text{R-078 safe packet} \\ &\rightarrow \text{R-079 full decomposition.} \end{aligned} \quad (10.1)$$

The decisions recorded at this checkpoint are:

- **Advanced:** full safe-packet reconstruction by the complete Hilbert current, with all companions retained.
- **Advanced:** spatially weighted Cameron–Martin square-function one-use and predictable base-current heat projection.
- **Failed:** square-function Cauchy at critical regularity, by the exact deficit $-4/15$.
- **Failed:** replacement of expected energy and sextic budgets by a predictable BMO/Carleson norm, by the rare-branch fixture.

- **Failed:** generic spatial gain from an unweighted bracket, by the opposite-carrier fixture.
- **Failed:** universal adapted Wick/carré-du-champ positivity, even with the exact positive square and trace.
- **Parked with a quantitative resume condition:** the positive gain route (9.1), which becomes subcritical for every $\gamma > 0$.

11 Devil’s-advocate self-test

1. **“The cross term $\langle f_j, i_j \rangle$ was lost.” DISMISSED.** It is contained in $\langle d_j \mathcal{J}_j, i_j \rangle$ because $d_j \mathcal{J}_j = d_j \mathcal{J}_0 + f_j$; equation (3.7) verifies the polarization literally.
2. **“The low term is already the R-066 finite-low endpoint.” UPHELD as false.** The complete low endpoint is (4.6), whereas $\mathfrak{L}_\ell$ contains $P_\ell \mathcal{J}_*$. Both remain.
3. **“R-077 fresh-packet cancellation kills $\mathfrak{F}_j$.” UPHELD as unproved.** The middle current changes with j , so f_j is not the difference sequence of one fixed terminal random variable.
4. **“The innovation is only a future coefficient.” UPHELD as false.** The derivative-feedback term on the second line of (5.2) is exact and generally nonzero.
5. **“The positive square makes the trace harmless.” UPHELD as false.** Equations (8.8)–(8.9) give a bounded smooth positive-frame counterexample.
6. **“The weighted control square function proves the packet bound.” UPHELD as false.** Cauchy yields the exact negative slack $-4/15$, and (7.8) rules out generic spatial repair.
7. **“Expected Boué–Dupuis budgets imply the required Carleson norm.” UPHELD as false.** The two expected budgets in (7.5) stay one while the conditional energy in (7.6) grows as N^6 .
8. **“R-063 already supplies $\gamma = 3/10$ in (9.1).” UPHELD as false.** R-063 reconstructs deterministic translated forests; later adapted controls require a new theorem.
9. **“This closes controlled-shell one-use, Nelson, or Sector A.” UPHELD as an overclaim.** The production near/far lower bound and the conditional low-current estimate remain open; the tier stays T4.

12 Reproduction and independent audit

The primary executable checks a coherent finite-tree current and coefficient frame, the low/injected/future trace split, the safe causal telescope, the future derivative commutator, all exact rational ledgers, the Hermite forest, the bounded Gaussian fixture, the weighted control square function, the rare-branch obstruction, and the base-current heat identity. The independent executable uses a two-component matrix frame, four probability roots, a different covariance split, four-node Gaussian quadrature, high-order Gauss–Hermite quadrature, and separately derived fractions.

Run from the repository root:

```
python codes/foundations/
a13_classii_full_safe_packet_frame_current_doob_verify.py
```

The child contracts are primary 51/51 and non-importing independent 42/42. The integrated verifier additionally pins the direct R-063, R-066, R-070, R-073, R-077, and R-078 authorities; checks source, note, PDF, manifest, run, negative-result, roadmap, gate, and claim surfaces; and fails closed if any lower-bound or Sector-A flag is promoted. The two finite-tree paid functionals are deliberately generic nonzero structural fixtures: they test the sign and telescope of $\mathcal{W} - P$. The production identification $P = N_{3,\text{nr}} + \hat{T}_{\leq}$ is the analytic substitution (4.1)–(4.4), pinned directly to R-078; the synthetic fixtures do not claim to reconstruct those production tensors numerically.

Result footer

Result ID: A13-CLASSII-FULL-SAFE-PACKET-FRAME-CURRENT-DOOB-
DECOMPOSITION

Precise statement:	The complete canonical R-078 safe packet has the exact expectation identity (4.4): low conditional current plus injected-current and future-feedback innovation blocks, their positive squares and Wick traces, minus the paid endpoint difference. The control coordinate has the weighted one-use square function (6.3), and predictable base currents obey the heat projection (6.7). Generic weighted-Doob shortcuts fail; the production lower bound is open.
Scope:	Finite cutoff; fixed positive floor; deterministic QII and covariance increments; common even regulator; strict-past one-shot shell controls; exact products; complete R-063/R-066 companions.
Dependencies:	A1; R-063; R-066; R-070; R-073; R-077; R-078.
Evidence grade:	ANALYTIC, EXACT, EXECUTED.
Reproduction command:	python codes/foundations/ a13_classii_full_safe_packet_frame_current_doob_ verify.py
Expected output:	Primary 51/51; independent 42/42; manifest-pinned integrated and aggregate PASS; exit zero.
Falsification gate:	Failure of (3.5), (4.4), (5.2), (6.3), or (6.7); omission of square, trace, forest, derivative feedback, paid subtraction, or either low object; exponent-ledger disagreement; or evidence drift.
Tier before / after:	T4 / T4; no promotion.
No-overclaim statement:	No production weighted packet lower bound, complete packet lower bound, finite-energy extension, controlled-shell energy one-use, q=10/9 Nelson, measure theorem, removal limit, or Sector-A closure is proved.
Next required action:	Prove a production near/far current--trace estimate: far-root heat-tail gain plus weighted CM square function, and a near-root signed Schur or gamma>0 bound with the full forest and both low terms.